SIH 2026 · SIH26044 · MINISTRY OF AYUSH

SkillSetu

AI-Powered Early Industry Readiness & Academia–Industry Collaboration Platform

SKILL INTELLIGENCE ENGINE

Detect skill gaps before they become placement failures.

DEFINING SKILLSETU

Two Ideas, One Product

Official SIH26044 Requirement

Connect students, industries, academicians and institutions for skill development, internships and placements.

Overall platform

Mentor's USP

Detect employability / skill-gap risk early — before placement season arrives.

Central intelligence layer & USP

SkillSetu = Platform + Intelligence Layer

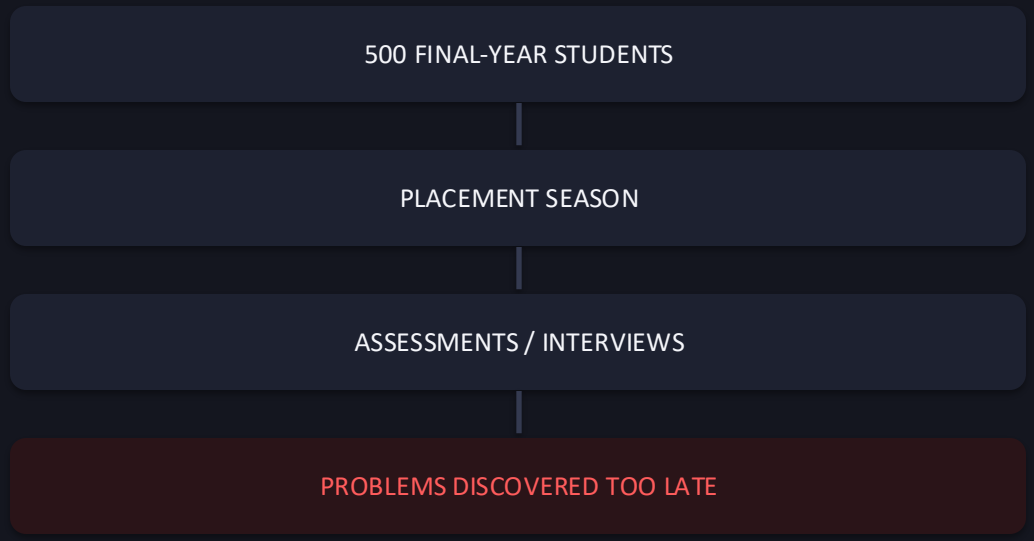
THE PROBLEM

The Real Problem

Institutions discover too late that their students are not industry-ready.

There is no continuous, data-driven early-warning mechanism for industry readiness.

A college may know marks, attendance, courses and certificates — but not whether students can demonstrate the competencies industry needs.



60% weak communication · 45% lack practical experience · 70% lack skills for emerging roles

Very little time left to intervene

THE SOLUTION

What We Are Solving

SkillSetu continuously answers four questions:

① **Where does each student currently stand?**

What skills does this student actually have?

② **What should they have?**

What competencies does industry require for their target role?

③ **What is missing?**

What are the student's specific skill gaps?

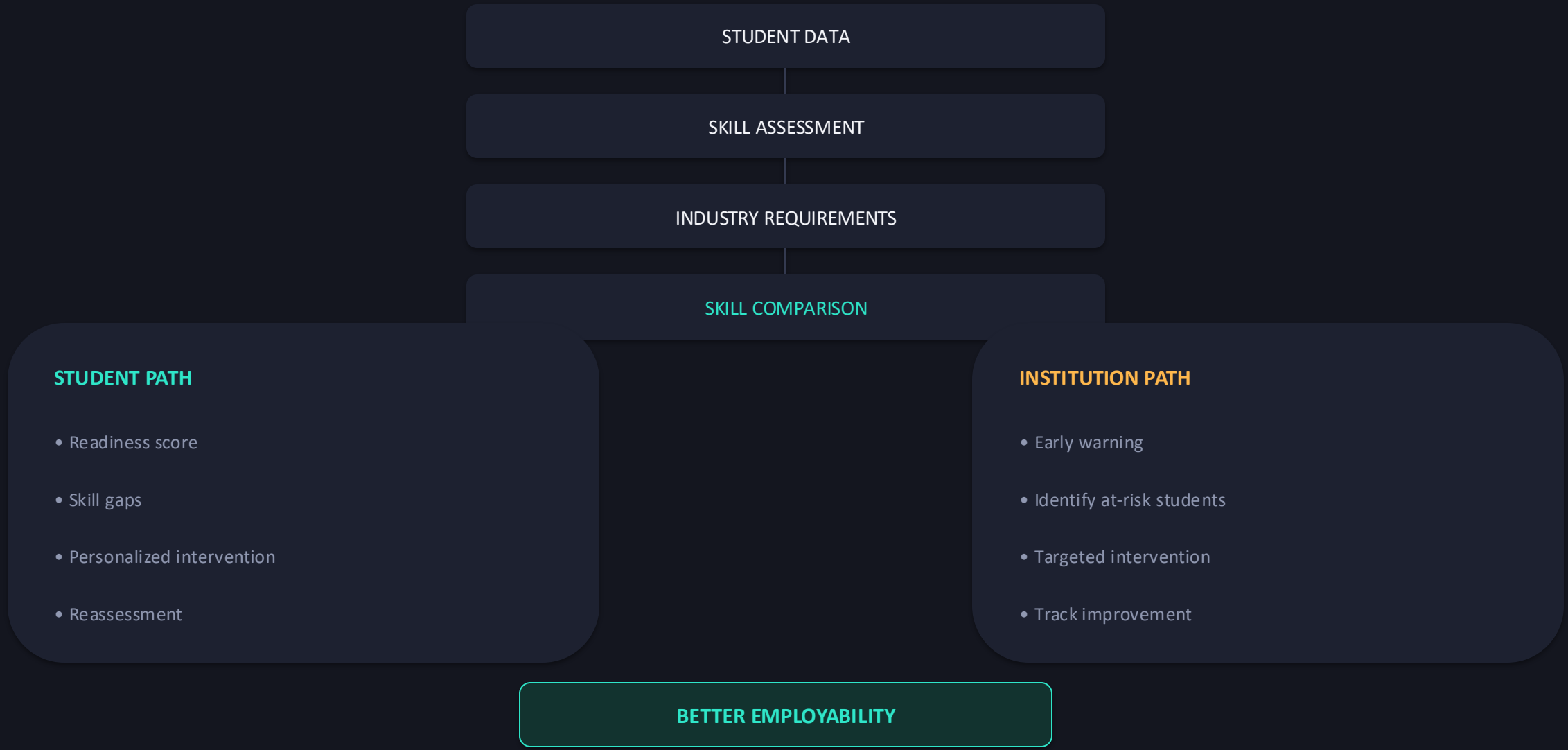
④ **What should happen next?**

What intervention should close those gaps?

This is the heart of the product: know → compare → detect → intervene → measure.

CORE LOOP

The Core Readiness Loop



ROLE-SPECIFIC READINESS

Industry Readiness Is Role-Specific

Student A

Target: Data Analyst

Evaluated against:

- SQL
- Excel / BI
- Statistics
- Python

Student B

Target: Cloud Engineer

Evaluated against:

- AWS
- Linux
- Networking
- Docker

Student C

Target: Frontend Developer

Evaluated against:

- React
- CSS
- JavaScript
- APIs

Same student ≠ same readiness. Target role changes the competency set.

SCORING MODEL

Two Scores, Not One Arbitrary Number

CORE EMPLOYABILITY

72%

MODERATE

Communication · Problem Solving · Teamwork
Practical Exposure · Portfolio · Interview Readiness

ROLE READINESS



Generally employable — significant Cloud Engineering gap.

INDUSTRY REQUIREMENTS

Companies Feed the Skill Intelligence Engine

Company post: Software Developer Intern

JavaScript · React · Node.js · REST APIs
SQL · Git · Problem Solving · Communication

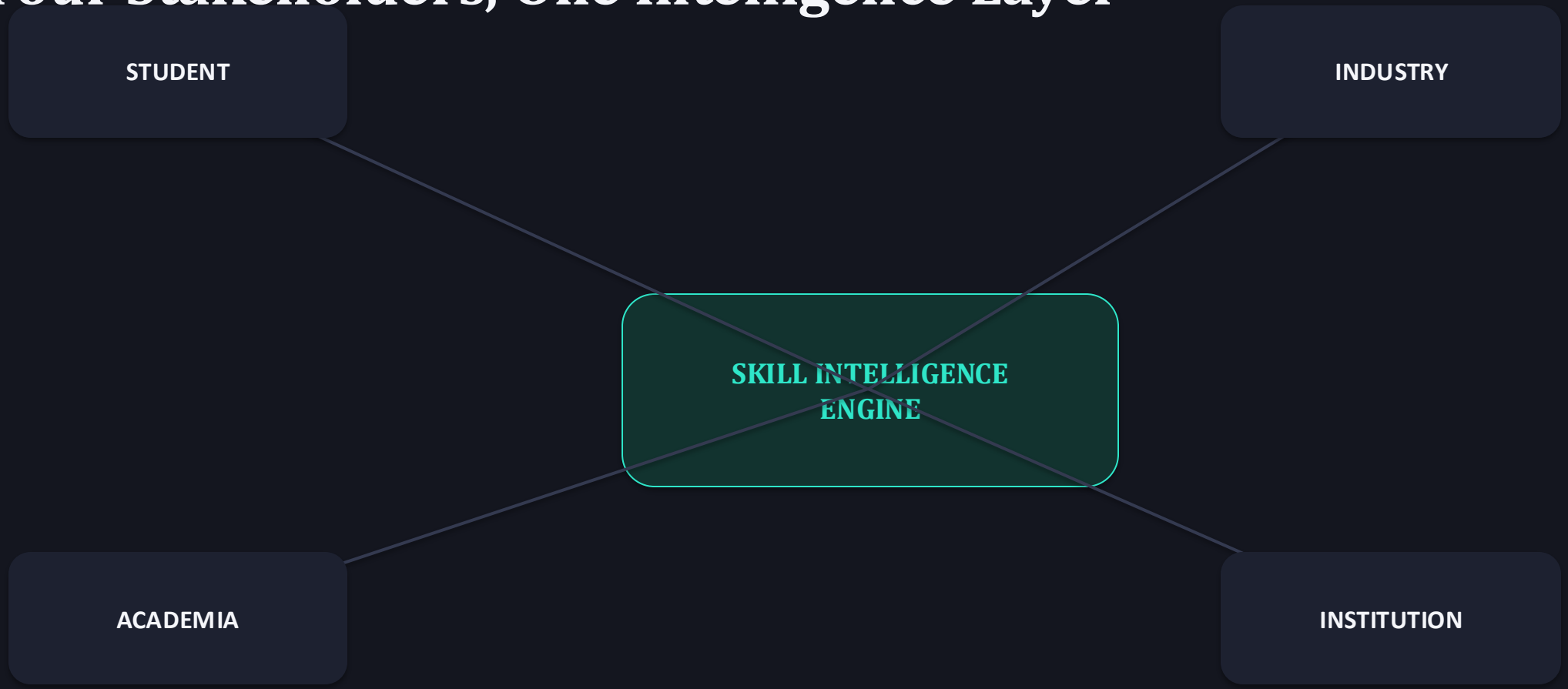
Aggregated across 50 companies



COMPANIES — REQUIRED SKILLS — SKILL DEMAND DATA — INTELLIGENCE ENGINE — INSTITUTIONS — TRAINING / CURRICULUM

PLATFORM ARCHITECTURE

Four Stakeholders, One Intelligence Layer



A common intelligence layer sits underneath all four portals.

STUDENT SIDE

Student Journey: Registration → Profile

1 **Registration**

College, degree, branch, year, graduation year + multiple career interests

2 **Build Profile**

Education · skills · experience · certifications · projects · GitHub · resume

3 **Skill Assessment**

Technical role-specific assessment + soft skills + aptitude where appropriate

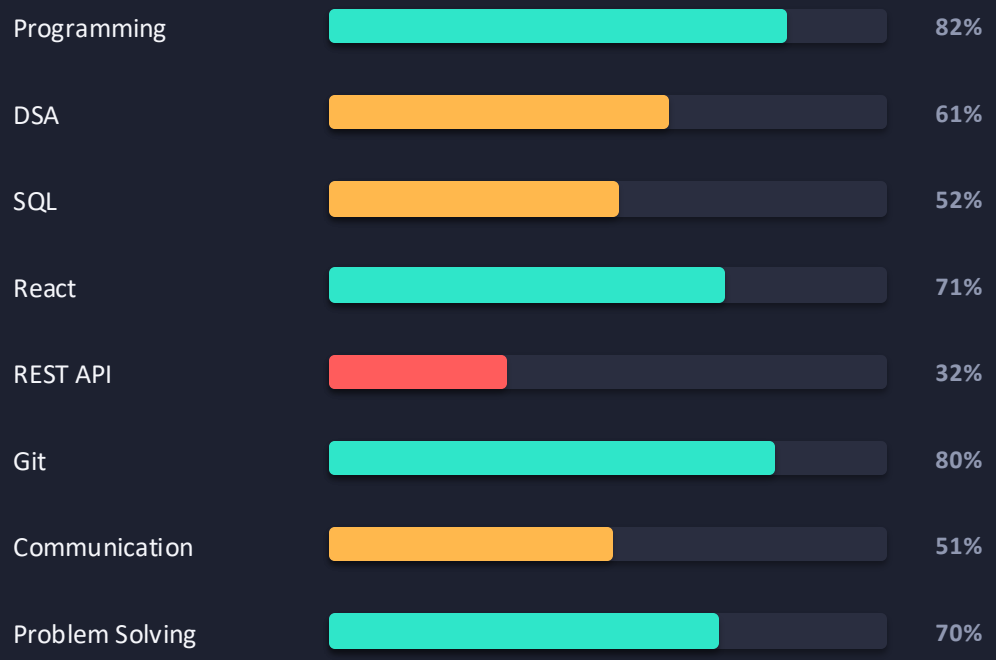
The assessment generates evidence for the student's competency profile.

Next → structured Skill Profile

STUDENT SIDE

Skill Profile: Turn a Resume Into Competency Evidence

STUDENT SKILL PROFILE



EVIDENCE SOURCES

- Technical assessments
- Soft-skill / aptitude assessments
- Projects and practical experience
- Certifications
- GitHub / portfolio
- Resume and verified records

Structured profile ≠ stored resume

INTELLIGENCE LAYER

Readiness Engine: Compare Student vs Industry

Competency	Required	Student	Gap
Python	70	82	Strong
SQL	75	52	Moderate gap
REST APIs	80	32	Critical
Git	60	80	Strong
Docker	50	20	Critical
Problem Solving	70	70	Strong

Example Role Readiness: 61% → 3 critical gaps identified

The engine is explainable because each score can be traced to defined competencies, evidence and weights.

EARLY-WARNING USP

Early Warning: Act Before Placement Season

🔴 AT RISK FOR BACKEND DEVELOPER

Your current profile does not meet 3 critical competencies commonly required for your target role.

REST APIs | Docker | SQL

You still have time to improve.

OLD MODEL

Placement season
↓
Assessment
↓
Gap discovered

NEW MODEL

Continuous data
↓
Early warning
↓
Intervention
↓
Reassessment

INTERVENTION ENGINE

Personalized Intervention: From Gap to Action

DETECTED GAP

SQL — 52%
REST API — 32%
Docker — 20%

Role: Backend Developer
Readiness: 61%

INTERVENTION PLAN

Week 1	SQL fundamentals
Week 2	Joins + subqueries
Week 3	Database project
Week 4	REST API + SQL integration
Week 5	Assessment

Possible interventions: learning programs · workshops · certifications · projects · mentorship · practice assessments

CONTINUOUS CYCLE

Reassessment: Measure Improvement

SQL

52% → 78%

AFTER INTERVENTION

Role Readiness

61% → 74%

AFTER INTERVENTION

+13%

READINESS IMPROVEMENT

Previous: 61%
Current: 74%

The platform proves whether an intervention actually improved readiness.

OPPORTUNITIES

Internship & Job Matching

RECOMMENDED OPPORTUNITIES

Backend Developer Intern

Company A

88%

MATCH

Frontend Intern

Company B

93%

MATCH

WHY THE MATCH?

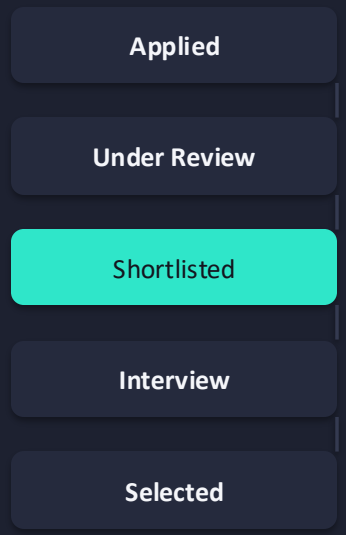
- ✓ Python
- ✓ Git
- ✓ SQL
- ✓ Problem Solving
- ⚠ Docker

View → Apply → Track

STUDENT SIDE

Application Tracking & Digital Portfolio

APPLICATION TRACKING



VERIFIED DIGITAL PORTFOLIO

- Skills + assessment results
- Certifications
- Projects + internships
- Achievements
- Verified experiences
- Shareable SkillSetu profile

Beyond a random PDF resume: one evolving evidence-backed profile.

INDUSTRY SIDE

Company Journey

1 · Register

Organization details + industry + verification + authorized recruiter

2 · Create Opportunity

Role + eligibility + technical skills + proficiency + soft skills

3 · Discover Candidates

Ranked by skill match with explainable evidence

4 · Shortlist & Engage

Shortlist · reject · contact · schedule interview · update status

Industry can also publish workshops, training, certifications, mentorship, FDPs and live projects.

INDUSTRY SIDE

Skill-Based Candidate Discovery

RECOMMENDED CANDIDATES

#1	Student A	91%
#2	Student B	87%
#3	Student C	84%
#4	Student D	81%

WHY IS STUDENT A 91%?

- ✓ Python
- ✓ SQL
- ✓ REST API
- ✓ Git
- ⚠ Docker
- ✓ Communication

Explainable matching: the recruiter sees the evidence behind the score.

ACADEMIA SIDE

Academician Journey

FACULTY PROFILE

- Department / expertise
- Research interests
- Skills and experience
- Projects / publications
- Industry interests

INDUSTRY COLLABORATION

- Faculty internships
- Industrial training
- FDPs
- Consultancy
- Collaborative research
- Industry projects
- Mentorship
- Guest lectures / workshops

Academician ≠ institution admin: this portal supports the individual faculty member.

INSTITUTION SIDE

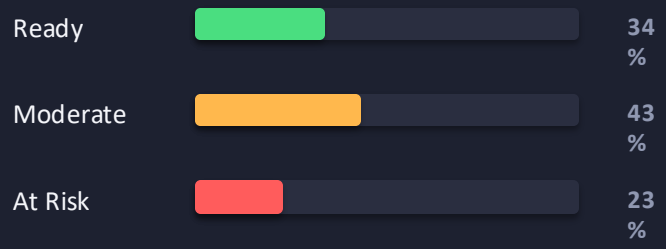
Institution Dashboard: Early Warning at Scale

OVERALL INDUSTRY READINESS

68%

MODERATE

Risk distribution



DRILL-DOWN: AT-RISK STUDENTS

Software Development	420 students
Data Analytics	280 students
Cloud	190 students
AI/ML	160 students
Other	100 students

Click a role → see its dominant skill gaps → create targeted intervention.

INSTITUTION SIDE

Institutional Intervention

CLOUD ENGINEERING

Top skill gaps

AWS	72%
Linux	61%
Docker	57%
Networking	48%
Cloud Security	41%

TARGETED PROGRAM

AWS Industry Readiness Bootcamp

Target: students with AWS readiness < 60%

- At-risk students
- AWS workshop
- Hands-on project
- Assessment
- Recalculate readiness

Instead of training everyone, target the students who specifically need the competency.

INSTITUTION + INDUSTRY

Industry Demand Dashboard

MOST DEMANDED SKILLS



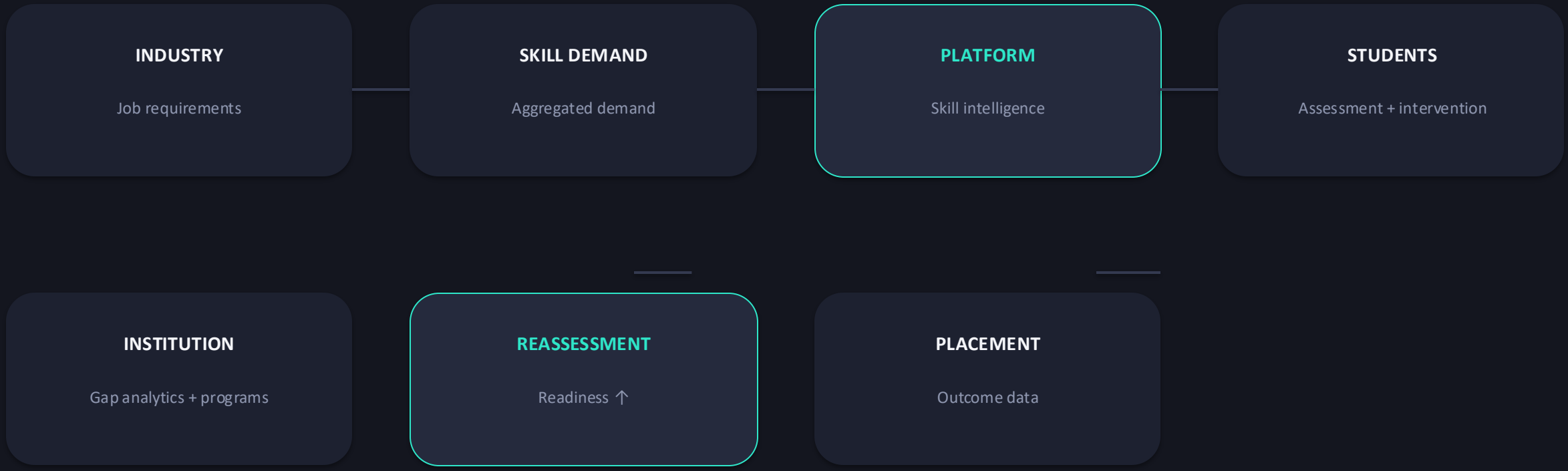
DEMAND TREND



Use demand data for curriculum planning, workshops, FDPs and placement preparation.

SYSTEM LOOP

The Complete Feedback Loop



Industry demand → student capability → targeted institutional action → measurable improvement → placement outcomes → new demand data.

AI COMPONENTS

Where AI Comes In

1	**Resume / Portfolio Understanding**	Extract skills and evidence from uploaded resumes and portfolios.
2	**Skill Normalization**	Map JS → JavaScript, React.js → React, and similar variants to a common taxonomy.
3	**Skill-Gap Explanation**	Explain why a competency is missing relative to the target role.
4	**Personalized Recommendations**	Generate a sensible intervention sequence from student + role + gaps.
5	**Opportunity Matching**	Add semantic similarity alongside structured eligibility and skill matching.

EXPLAINABILITY

Readiness Scoring Should Be Deterministic

DON'T SAY

“GPT gave the student 72%.”

That is difficult to defend, reproduce or audit.

DO SAY

Deterministic scoring engine

- Technical competencies
- Soft skills
- Practical experience
- Assessment evidence
- Eligibility
- Defined weights / thresholds

AI = understanding + normalization + explanation + recommendations.

PRODUCT STRUCTURE

Complete Website Structure

LANDING / AUTH

- Landing Page
- Login / Register

STUDENT PORTAL

- Dashboard
- Profile
- Skill Assessment
- Skill Profile
- Readiness
- Skill Gaps
- Learning / Interventions
- Internships / Jobs
- Applications
- Portfolio
- Mentorship

INDUSTRY PORTAL

- Dashboard
- Company Profile
- Post Internship / Job
- Training Programs
- Candidate Matching
- Shortlisting
- Applications
- Collaboration

ACADEMIA PORTAL

- Dashboard
- Faculty Profile
- Faculty Internships
- FDPs
- Consultancy
- Research
- Mentorship
- Industry Collaboration

INSTITUTION PORTAL

- Dashboard
- Student Readiness
- At-Risk Students
- Skill Gap Analytics
- Industry Demand
- Interventions
- Internship Analytics
- Placement Analytics
- Reports

PRODUCT VISION

What Happens When Everything Connects?

STUDENT

What do I need to improve?

COMPANY

Who has the skills I need?

FACULTY

Where can I collaborate with industry?

INSTITUTION

Which students are at risk, why, and what should we do?

SKILLSETU

Connect all four through a shared Skill Intelligence Layer.

POSITIONING

The Central USP

“SkillSetu helps institutions identify and intervene in student employability risks early — by continuously mapping student competencies against real industry requirements, rather than discovering skill gaps during placement season.”

Secondary: Connect students to personalized learning and relevant opportunities while giving industries access to skill-matched talent.

SIH ALIGNMENT

How SkillSetu Maps Back to SIH26044

Skill Management	Student-Industry Collaboration
Student assessments	Competency profile
Readiness engine	Intervention engine
Role readiness	Opportunity portal
Skill matching engine	Learning programs
Academician portal	Collaboration module
Verified student profile	Early-warning dashboard
Skill-demand analytics	4 stakeholder portals
Resume / certificates / records	Projects / FDPs / mentorship / research

FINAL POSITIONING

The Most Important Conceptual Change

NOT JUST

“A platform that helps students find opportunities.”

INSTEAD

“A continuous Industry Readiness Intelligence System that detects skill gaps early, recommends interventions, measures improvement, and then connects students with relevant industry opportunities.”

TEAM DIRECTION

What the Team Should Build Around

The product architecture should revolve around one question:

“Can we identify an employability risk early enough to do something about it?”

DETECT

Measure current capability

COMPARE

Map against industry demand

INTERVENE

Recommend targeted action

MEASURE

Reassess improvement

CONNECT

Match to opportunities

This is the logic every major screen and database entity should support.

NEXT STEP

Next Step: Turn the Concept Into a Buildable MVP

PRODUCT SPECIFICATION	Exact pages, screens, buttons, navigation and user journeys.
DATA MODEL	Database entities, relationships, roles, evidence and opportunity records.
READINESS ENGINE	Competency taxonomy, weights, thresholds, scoring and explainability.
AI LAYER	Resume parsing, normalization, explanations, recommendations and semantic matching.
MVP FOR INTERNAL HACKATHON	A realistic vertical slice demonstrating early warning → intervention → reassessment.
DEMO STORY	One student, one target role, one institution dashboard and one company matching flow.

Concept → Product Specification → MVP → Internal Hackathon Prototype